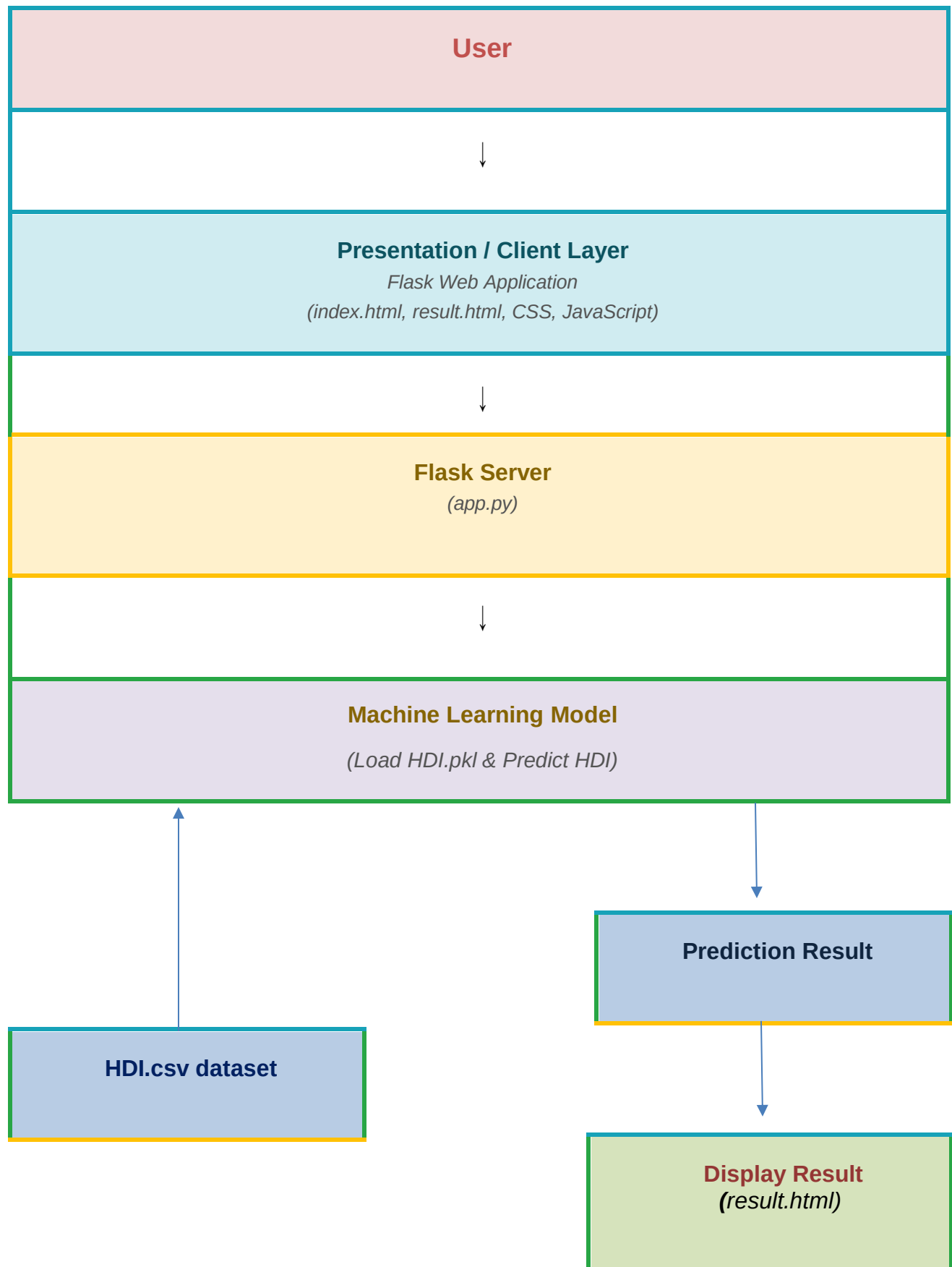


Solution Architecture

Date	15 March 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for entering development indicators and displaying the predicted HDI score and category.	HTML5, CSS3, JavaScript, Flask Templates
Application Layer (Flask Server)	Handles HTTP requests, validates user input, loads the trained model, performs predictions, and returns results to the frontend.	Python, Flask
Core Logic Service	Performs machine learning inference using the trained Linear Regression model to predict the Human Development Index.	Scikit-learn, NumPy, Pandas
Data Layer	Stores the HDI dataset used for training and the trained machine learning model used for prediction.	CSV File (HDI.csv), Pickle Model (HDI.pkl)

Technologies Used

- Python
- Flask
- Scikit-learn
- Pandas
- NumPy
- HTML5
- CSS3
- JavaScript
- Jupyter Notebook
- Visual Studio Code